بسم الله الرحمن الرحيم

Diagnosis and physiology of Pregnancy

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Objectives

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1. Identify the signs and symptoms of pregnancy
2. Recognize the adaptation of body system in pregnancy.
3. Identify the Hormonal Changes in Pregnancy
4. Estimate date of delivery
5. Identify Nursing Care with First Prenatal Visit
6. Prenatal Health History for identifying high risk women
7. Recognize the necessary lab. Investigation for pregnant women
8. Address common complaints that may arise during early pregnancy

Physiological changes of pregnancy

Adaptation of reproductive system	Adaptation of other system
■Uterus ■Cervix ■Ovaries, fallopian tubes ■Vagina ■Vulva ■Breasts	■Cardiovascular ■Respiratory ■Urinary ■Metabolic changes ■Indocrine ■Skeletal

1- Reproductive system

Uterus

- Increase in size, weight 50- 1000 gm.
- Globular .
- Braxton Hichs (painless, irrgular contractions.
- Formation of decidua
  - Basalis : uderlying the ovum
  - Capsularis : covering the ovum
  - Vera : covering the rest of uterine cavity .

Skin changes

- Chloasma or melasma gravidarum
- Striae
- Linea nigra

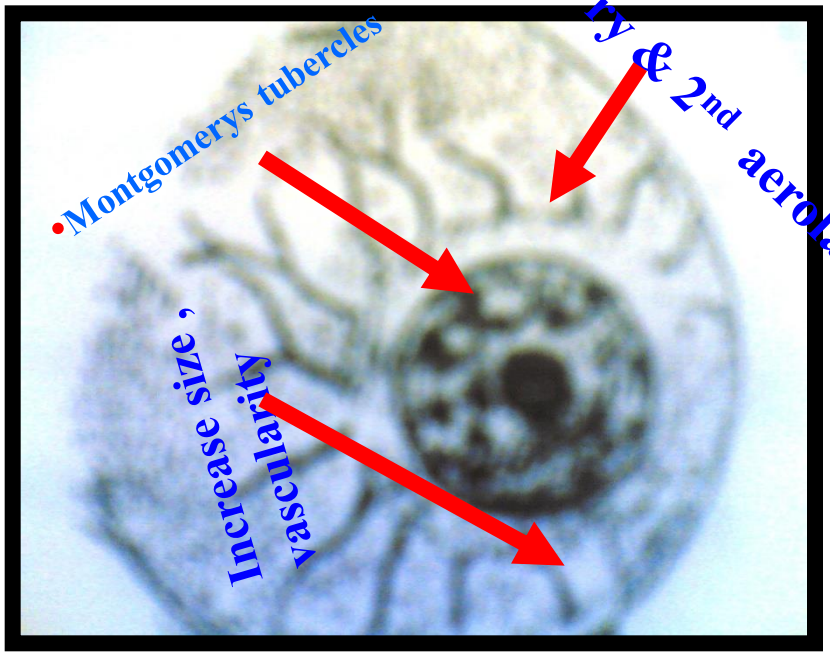
Chloasma

Melasma or Chloasma

- Also known as the mask of pregnancy
- More common in dark skin people
- More pronounced in the summer
- Fades a few months after delivery
- Repeated pregnancy can intensify
- Can occur in normal non-pregnant
- women with harmless hormonal
- imbalances or women on OCPs or depo

Striae

- Reddish slightly depressed
- Breasts, thighs, and abdomen
- In future pregnancies they appear as shiny, sil



Cardiovascular Changes

Heart rate	Cardiac output	Blood volume	Blood pressure	Systemic vascular resistance
Increase heart rate by 10-15 b/m	to 30% during the first 2nd trinester	40-50 % physiological anemia	* Slight ↓ with return to baseline by 3rd trimester	

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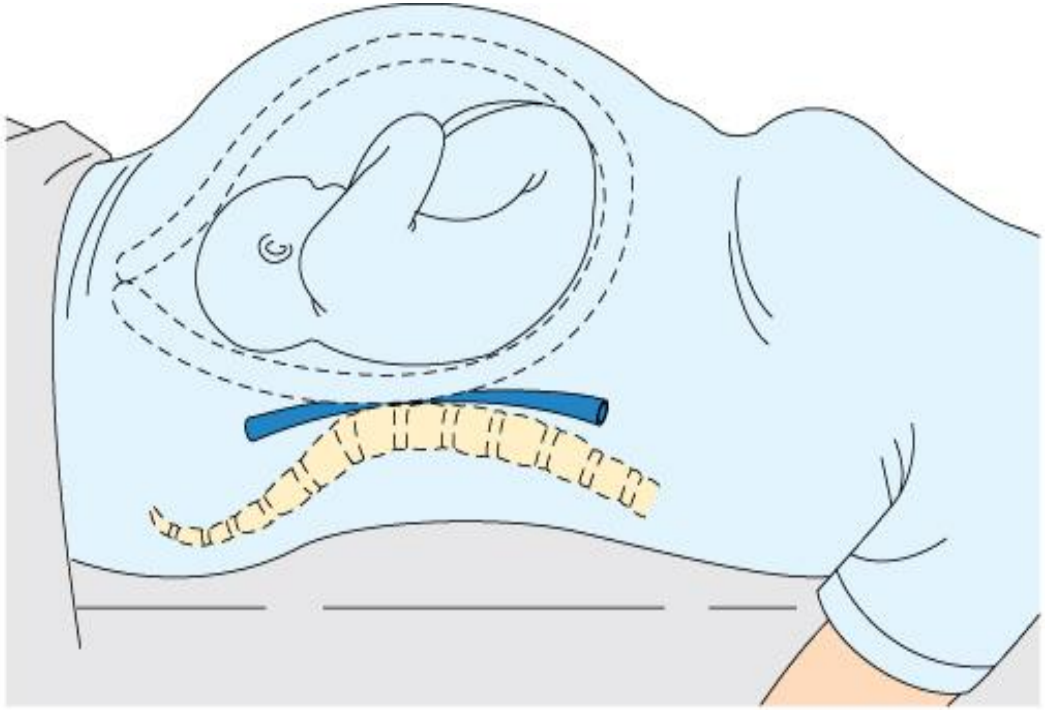
Glomerular filtration rate	Basal metabolic rate	Respiratory rate	Urine output	Skeletal
			No change	□**Lamber lordosis** □ **Softening of pelvic joint and ligaments**

Supine Hypotensive Syndrome



Hormonal Changes in Pregnancy

Hormone		Functions
Estrogen		1. Increase uterine muscle mass 2. Increase blood flow to uterus 3. Prepare breasts for lactation
Progesterone		1. Relax venous walls 2. Inhibit uterine contractions
Human chorionic gonadotropin (hCG)		1. Stimulate estrogen/progesterone 2. production
Relaxin		1. Stimulate estrogen/progesterone 2. production
Human placental lactogen		1. Maturation of breast ducts/alveoli 2. Stimulate lactation
Prolactin		1. Insulin antagonist 2. Allow adequate glucose for fetal demand

Estimated Date of Delivery

- **Establishing an accurate date of delivery is important to:**
  - Determine timing of antenatal screening
  - Monitor growth of the fetus
  - Detect timing of delivery
- **Common abbreviations denoting delivery date are:**
  - **EDD.** ..... Estimated date of delivery
  - **EDC.** ..... Estimated date of confinement
  - **EDB.** ..... Estimated date of birth

Estimate date of delivery

(Naegele's Rule)

- Formula used to estimate date of delivery
- Count back 3 months and add 7 days to the last normal menstrual period (LNMP) reported by the patient

Example: The patient states that her LNMP was April 20th

April is the	4th month	20th day
	−3 months	+ 7 days
	1st month	27th day

The baby is estimated to be due on January 27th of the following year

Trimesters of Pregnancy

Normally, pregnancy continues for 40 weeks or 280 days

- 1st trimester ------ conception until 12 weeks' gestation
- 2nd trimester --------- 13 weeks until 27 weeks' gestation
- 3rd trimester---------- 28 weeks until 40 weeks' gestation

Early prenatal care

Schedule of Prenatal Visits (low-risk pregnancy)

- Monthly until 28 weeks' gestation
- Biweekly from 28 weeks until 36 weeks
- Weekly from 36 weeks until delivery

Prenatal Health History

Performing a thorough health history in the prenatal period is essential to planning nursing care and identifying high-risk women.

- Medical history
  - Chronic illness
  - Current and recent medication
  - Recent acute illness
  - Childhood illnesses
- Surgical history
  - Problems with anesthesia
  - Previous surgeries
  - Uterine/cervical surgeries
- Obstetrical history
  - Type of deliveries: vaginal/cesarean
  - Complications with past pregnancies
  - Infertility
  - Documentation of obstetrical history

Descriptive Term	Definition
Gravida (G)	Number of pregnancies
Term (T)	Number of deliveries after 37 weeks
Preterm (P)	Number of deliveries after 20 weeks but before 38 weeks
Abortion (A)	Number of deliveries before 20 weeks, either spontaneous or induced
Living (L)	Number of living children

- Sexual history
  - Number of sexual partners
  - Sexually transmitted infections
  - Sexual abuse
  - Methods of contraception
  - Condom use
- Social history
  - Use of recreational drugs
  - Smoking
  - Domestic abuse
  - Educational level/ability to read
  - Economic status
  - Type of health insurance
  - Need for community referrals
    - Transportation
    - Nutrition
    - Medications

Nursing Care with First Prenatal Visit

- Determine EDD based on LNMP
- Document current gestational age (gestational wheel is a tool for quick reference to current gestational age)
- Document baseline vital signs
- Document height, weight, and body mass index (BMI)
- Obtain urine specimen and test for presence of:

Substance	Expected Finding
Glucose	Negative/Trace
Protein	Negative/Trace

- Auscultate fetal heart tones
- Measure fundal height in centimeters from symphysis pubis to the top of the fundus
  - Uterine size increases in pregnancy in a predictable pattern and is measured to gauge fetal growth
  - Fundal height that is lagging or greater than expected should be further investigated

Weeks' Gestation	Fundal Height
12	Just above symphysis pubis
16	Halfway between symphysis pubis and the umbilicus
20	At the umbilicus
21–36	Fundal height generally matches weeks gestation in centimeters EXAMPLE: Fundal height at 28 weeks should be approximately 28 cm.

- Provide appropriate education for gestational age
- Discuss procedure for lab testing

Laboratory Tests

Laboratory Tests

Common Laboratory Tests	Expected Finding in Pregnancy
HIV *Check state laws regarding HIV testing in pregnancy	Negative
Blood type	A, B, AB, O
Rh factor	Negative/Positive
Antibody screen	Negative
Hemoglobin	>11.5 mg/dL
Hematocrit	>33%
Platelets	150,000–400,000 mm^3
WBC	5,000–12,000 mm^3
RPR	Negative
Hepatitis B antigen	Negative
Rubella titer	1:8 Immune
Hemoglobin electrophoresis	AA, unaffected
Chlamydia culture	Negative
Gonorrhea culture	Negative
Pap smear	Normal cytology

Education in the Early Prenatal Period

- Elevated estrogen and progesterone levels in early pregnancy generate changes in the body, causing pregnancy associated discomforts
- Offer suggestions to lessen discomforts
- Teach patient to report symptoms that may indicate a potential complication (in red)

Discomfort	Patient Education
Urinary frequency	Related to uterine position/weight Encourage frequent emptying of bladder Discourage limiting oral fluids Report burning or pain with urination
Nausea and vomiting	Related to elevated hormone levels Encourage small, frequent meals Eat crackers before rising Avoid pungent odors, spicy or greasy food Discuss limited time frame for nausea (subsides around 12 weeks' gestation) Report excessive vomiting

Education in the Early Prenatal Period *(Cont'd)*

Discomfort	Patient Education
Emotional lability	Related to hormone changes Discuss normalcy of emotional changes with patient and partner Ambivalence normal in first trimester Report constant crying, inability to care for self, suicidal thoughts
Leukorrhea	Related to vasocongestion of mucous membranes Avoid tampon use and douching Wear peri-pad to absorb discharge Encourage cotton underwear Report vaginal discharge with an odor or color, vaginal bleeding, or leaking of amniotic fluid
Breast discomfort	Hormone-related breast development often first presumptive sign of pregnancy Wear a supportive bra Colostrum may be expressed in pregnancy Introduce the value of breastfeeding Introduce/reinforce breast self-exam Report any breast lump or unusual discharge
Fatigue	Related to rapid hemodynamic and metabolic changes in the first trimester Encourage naps during the day Encourage prenatal vitamins Encourage healthy diet Report syncope and vertigo
Nasal stuffiness/ epistaxis	Related to vasocongestion of mucous membranes Increased humidity in home may help Warm compresses to sinus area Avoid over-the-counter (OTC) cold remedies Report fever, green/yellow nasal discharge, or frequent nosebleeds

Thank You